

Aakash Thapa

Lubbock (Texas), United States | aakashthapa.work@gmail.com | +1 (806) 283-3741 | LinkedIn

Summary

M.S. in Computer Science candidate and Senior Full-Stack Software Engineer with 6+ years of experience building scalable web, mobile, and AI-powered applications using Python, React, Django, and AWS. Experienced in LLM-powered document platforms and AI-assisted development using Cursor for implementation, refactoring, testing, and debugging.

Education

Texas Tech University , M.S. in Computer Science (Expected) Edward E. Whitacre Jr. College of Engineering - Lubbock (Texas), United States	Aug 2026 – Dec 2027
Kurukshetra University , B.Tech in Computer Science - 71.64% E-Max Group of Institutions - Haryana, India	Aug 2015 – Jun 2019

Experience

Senior Full-Stack Software Engineer , Vivpro – Pune, India	Nov 2023 – Jul 2026
---	---------------------

- Architected an **event-driven** subscription reporting service generating **PDF and DOCX reports** for regulatory document updates on user-defined schedules using **SQS-triggered Lambda** containers hosted in **ECR**, **S3** storage, and CloudWatch monitoring.
- Engineered **LLM-powered document** services using **React and Flask**, including document Q&A, news briefs, and configurable summaries across the top 10–100 relevant documents, increasing user engagement by **40%**.
- Designed and deployed **automated DAG-based pipelines** that ingested **10,000–30,000** documents per source from 6+ global regulatory sources, stored content in **Amazon S3**, and indexed it in **AWS Kendra**, reducing manual ingestion effort by over **50%**.
- Developed and maintained a **notification service**, delivering **daily email and web alerts** on document updates to **700+** users, increasing retention by **35%**.
- Championed team adoption of **Cursor** and **LLM-assisted engineering workflows** through live demonstrations and practical guidance, accelerating feature delivery by approximately **35%**.
- Migrated the entire frontend from **React 17 to React 18**, upgraded **TypeScript** and supporting libraries, resolved breaking changes, and upgraded AWS Amplify authentication modules using Amazon Cognito.
- Developed a reusable filter module with **virtualized infinite scrolling**, reducing frontend development effort by **80%** and enabling reuse across **10+** core sections.
- Reduced reporting load times by **30–50%** by optimizing **backend queries**, implementing **Redis caching**, and enforcing client-specific **API rate limits**.
- Implemented scalable full-text search across 30,000+ documents using **Elasticsearch** and **AWS Kendra** while maintaining **99.9%** availability.
- Participated in production **on-call rotations**, resolving incidents and maintaining a **mean time to recovery of under 2 hours**.

Full-Stack Software Engineer II , Fixcraft – Gurgaon, India	Aug 2021 – Jul 2023
--	---------------------

- Implemented **Django-based microservices** for WhatsApp, email, SMS, and in-app **notifications** and **integrated Razorpay payments**.
- Built an **e-commerce website** from the ground up using **Python, Django, and React**, catering to both **B2B and B2C** demands for automobile spare parts.
- Led end-to-end development of **React Native** applications for **Android** and **iOS**, increasing the customer base by **28%** and migrating web users to the mobile applications.
- Migrated a **React** website to **Next.js** with server-side rendering, increasing its Lighthouse SEO score by **30%**.
- Created a **private npm package** that improved code reuse and reduced new-feature rollout time by **35%**.
- Guided a team of **3 front-end developers**, managing task prioritization and facilitating daily Scrum meetings.

Team Lead, Innostax Software Labs – Gurgaon, India

Sep 2020 – Aug 2021

- Devised front-end architecture for **3+ projects** using **React** with **responsive design** and reusable components, boosting user engagement by **16%** and reducing bounce rates by **13%**.
- Engineered backend structure for **2+ projects**, establishing databases and relationships via **Node.js**, **Express.js**, and **PostgreSQL**.
- Managed a team of **2–4** engineers, supporting hiring, task allocation, technical reviews, and on-time delivery across multiple client projects.
- Led **code reviews**, offering constructive feedback and expediting new member training.

Full-Stack Software Engineer I, Innostax Software Labs – Gurgaon, India

Aug 2019 – Sep 2020

- Built frontend features and component libraries using **React and Redux** for **2+** client projects, reducing duplicate UI work and accelerating page delivery.
- Created and optimized database tables, **REST APIs**, and **query performance** (including **index creation**), enhancing application speed and efficiency.
- Implemented unit Testing and end-to-end Testing using frameworks such as **Jest**, **Mocha**, and **Playwright** to improve code reliability.

Technologies

Languages: JavaScript, TypeScript, Python, HTML, CSS, C++

Frontend: React, Next.js, React Native, Redux

Backend: Django, Flask, FastAPI, Node.js, Express.js, REST APIs, Celery

Data & Messaging: PostgreSQL, Redis, Elasticsearch, Kafka

Cloud & DevOps: AWS (EC2, S3, SQS, Lambda, ECR, Kendra, Cloudwatch), Docker, Kubernetes, Git, GitHub, UNIX/Linux

AI Tools: Cursor, LLMs (GPT-4, Claude), Prompt Engineering, Claude Skills

Core Concepts: Data Structures and Algorithms, System Design, Object-Oriented Programming

Testing & Observability: Datadog, Jest, Mocha, Playwright, Selenium, SonarCloud, Sentry